



Bahir Dar Institute of Technology
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Final year Project Titles

- *An Intelligent Drug Tracking, Expiration Monitoring, and Regulatory Oversight System*
- *AI-Powered Community Issue Reporting and Resolution Tracking System*
- *AI-Powered Legal Assistant for Ethiopian Labor and Proclamation Advisory*

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TITLE 1: An Intelligent Drug Tracking, Expiration Monitoring, and Regulatory Oversight System

1. Introduction

The pharmaceutical sector plays a very important role in protecting public health in Ethiopia. It operates under regulatory institutions such as the Ethiopian Pharmaceutical Association and the Ethiopian Food and Drug Authority. However, many pharmacies still rely on manual or semi-manual systems to manage drug inventory, track expiration dates, and prepare regulatory reports.

Pharmacists often spend a significant amount of time recording drug information, checking expiry dates, and preparing reports manually. At the same time, regulatory authorities do not have a centralized system to monitor national drug inventory in real time. Citizens also face difficulties when searching for specific medicines, especially when they are out of stock at nearby pharmacies.

To address these issues, this project proposes a digital system that:

- Provides a user-friendly public portal with map integration so citizens can easily find pharmacies that have a required medication in stock.
- Offers role-based dashboards that give regulatory bodies real-time visibility into pharmacy operations.
- Automates data recording and report generation to reduce workload for pharmacists.

2. Problem Statement

The current pharmacy management system in Ethiopia faces several challenges that affect public safety, regulatory performance, and daily pharmacy operations:

- **Lack of Centralized Visibility:** Regulatory bodies do not have real-time access to national drug inventory, distribution patterns, or compliance status.
- **Manual Expiration Tracking:** Many pharmacies track expiration dates manually or using basic spreadsheets, increasing the risk of selling expired medications.
- **Inefficient Reporting:** Reporting to regulatory authorities is mostly paper-based and time-consuming, often leading to delays and inconsistencies.
- **Weak Recall Management:** When a drug recall occurs, there is no fast and reliable way to identify and notify all pharmacies that have the affected batch.
- **Difficult Access for Citizens:** Patients may waste time and money visiting multiple pharmacies to search for specific medicines.
- **Financial Losses:** Pharmacies lose money due to expired stock that could have been managed earlier with proper alerts.
- **Inconsistent Data Management:** Data recording methods vary from one pharmacy to another, and compliance monitoring depends mainly on manual inspections.

These issues reduce transparency, weaken regulatory control, and may negatively affect public health safety.

3. Project Objectives

General Objective

To design and implement a centralized, cloud-based, real-time pharmacy inventory and drug expiration monitoring system that connects registered pharmacies with national regulatory authorities in Ethiopia.

Specific Objectives

The system aims to:

- Provide real-time inventory tracking for registered pharmacies.
- Automate expiration monitoring at the batch level.
- Generate alerts for near-expiry and expired drugs.
- Enable centralized monitoring and data analysis for regulatory authorities.
- Simplify regulatory reporting and automate compliance processes.
- Support a digital drug recall mechanism at the national level.
- Offer a public portal where citizens can search for drug availability by location.
- Improve overall efficiency, transparency, and data-driven decision-making.

4. Proposed System Overview

The proposed solution is a secure, cloud-based digital platform designed to modernize pharmacy inventory management and drug expiration monitoring in Ethiopia.

The system connects three main user groups through separate interfaces:

- **Pharmacy Dashboard** – For managing inventory and daily operations.
- **Regulatory Dashboards** – For the Ethiopian Pharmaceutical Association and the Ethiopian Food and Drug Authority.
- **Public Portal** – For citizens to search for medicines.

The platform will use role-based access control to ensure security and organized usage. An intelligent component will operate in the background to:

- Automate routine processes,
- Provide predictive insights such as potential shortages,
- Support better decision-making for pharmacies and regulators.

5. Technology Stack

Frontend

- React.js – For building interactive and responsive user interfaces
- Tailwind CSS – For modern and user-friendly design
- Redux Toolkit – For managing application state
- React Router – For smooth navigation between dashboards
- Axios – For API communication
- Chart.js or Recharts – For displaying analytics and reports

Backend

- Node.js with Express.js – For developing RESTful APIs
- MongoDB – For storing pharmacy data, drug batches, and user records
- Mongoose – For data modeling and validation
- JWT (JSON Web Tokens) – For secure authentication and authorization
- Socket.io – For real-time updates and notifications

6. System Overview, Architecture, and Impact

This project introduces a centralized digital platform that connects pharmacies, regulatory authorities, and the public into one secure system for managing drug inventory, expiration tracking, compliance, and public access.

Pharmacy Dashboard

Pharmacies will log in securely to:

- Register drugs with batch number, expiry date, quantity, and price
- Update stock in real time
- Automatically deduct stock upon sale
- Track drugs at batch level
- Receive alerts for safe, near-expiry, and expired drugs

Expired drugs will automatically be restricted from sale. The system will also support:

- Barcode scanning
- Bulk data upload
- SMS or email notifications
- Automatic report generation for stock, sales, expiry, and compliance

Regulatory Dashboards

Regulatory authorities will be able to:

- Monitor national pharmacy inventory in real time
- Track batch-level information
- Manage drug recalls efficiently
- Verify proper disposal of expired drugs
- Analyze compliance and performance
- Access audit logs and inspection reports

Public Drug Search Portal

Citizens will be able to:

- Search for medicines by name
- View available pharmacies
- Check prices and locations
- Reserve medications

This will reduce delays and improve access to essential medicines.

7. Advanced Features and Technology

The system may include:

- Shortage prediction using intelligent analysis
- Recommendations for redistributing near-expiry drugs
- Automated blocking of recalled drugs

The system can be implemented using:

- Spring Boot for backend development
- WebSocket for real-time updates
- React for frontend development
- MySQL or PostgreSQL for structured data storage
- Secure cloud hosting with encrypted communication

8. Expected Impact

The proposed system is expected to:

- Reduce the circulation of expired drugs
- Improve regulatory monitoring and oversight

- Increase transparency within the pharmaceutical sector
- Minimize manual paperwork and reporting
- Enhance public trust
- Support data-driven healthcare management in Ethiopia

TITLE 2:AI-Powered Community Issue Reporting and Resolution Tracking System

1. Introduction

Urban communities in Ethiopia regularly face public service problems such as damaged roads, water leakage, waste accumulation, streetlight failures, blocked drainage systems, and other infrastructure-related issues. In many cases, reporting these problems depends on informal communication, manual paperwork, or physically visiting municipal offices. These traditional methods often result in delayed responses, poor tracking, and limited transparency.

Municipal authorities also face challenges in managing and prioritizing complaints effectively. Without a centralized and real-time system, it becomes difficult to monitor reported issues, allocate resources properly, and ensure accountability. On the other hand, citizens usually have no clear way to track the progress of their complaints or confirm whether any action has been taken.

This project proposes an AI-Powered Community Issue Reporting and Resolution Tracking System that aims to digitize and centralize the complaint management process. The system will provide a simple and user-friendly platform for citizens to report issues using descriptions, images, and geo-location, while offering intelligent dashboards that help municipal authorities manage, prioritize, and resolve cases more efficiently.

2. Problem Statement

The current system for managing community issues has several weaknesses:

- **Lack of a Centralized Reporting Platform:** There is no unified digital system where citizens can report public service problems.
- **Manual and Slow Processing:** Many complaints are handled manually, leading to delayed responses.
- **No Real-Time Tracking:** Citizens cannot monitor the progress of their submitted issues.
- **Poor Prioritization:** Municipal offices lack tools to automatically categorize and prioritize urgent issues.
- **Limited Transparency and Accountability:** There is no proper audit trail to track how complaints are handled.

- **Inefficient Resource Allocation:** Without proper data analysis, it is difficult to assign resources effectively.
- **Lack of Data-Driven Planning:** There is no structured database to analyze recurring problems or identify high-risk areas.

As a result, response time is slow, public trust decreases, and governance efficiency is reduced.

3. Project Objectives

General Objective

To design and implement a centralized, cloud-based, AI-powered community issue reporting and resolution tracking system that securely connects citizens and municipal authorities.

Specific Objectives

- Develop a user-friendly platform for citizens to report issues with photo and location support.
- Implement intelligent classification and prioritization of reported issues.
- Enable real-time tracking of issue status (Submitted → In Review → In Progress → Resolved).
- Provide role-based dashboards for different municipal departments.
- Generate automated notifications and alerts.
- Offer data analytics and performance insights for decision-makers.
- Improve transparency, accountability, and public trust.
- Support data-driven resource planning and infrastructure management.

4. Proposed System Overview

The proposed system is a smart digital platform designed to improve how community issues are reported and resolved.

The platform will serve three main user groups:

- **Citizens (Public Portal)**
- **Municipal Authorities (Department Dashboards)**
- **System Administrators**

The intelligent component of the system will:

- Automatically classify issues (road, water, sanitation, electricity, etc.)
- Detect urgency levels
- Recommend the appropriate department
- Identify hotspot areas based on recurring reports
- Provide basic predictive insights for planning purposes

Role-based access control will be implemented to ensure secure and authorized use of the system.

5. Technology Stack

Frontend

- React.js – For building an interactive and responsive user interface
- Tailwind CSS – For a clean and mobile-friendly design
- Redux Toolkit – For managing application state
- React Router – For smooth navigation
- Axios – For backend communication
- Recharts or Chart.js – For analytics dashboards
- Google Maps API or Leaflet – For geo-location integration

Backend

- Node.js with Express.js – For developing RESTful APIs
- MongoDB – For storing user data, issue reports, and logs
- Mongoose – For data modeling and validation
- JWT – For secure authentication and role-based access control
- Socket.io – For real-time notifications
- AI integration (via API or ML service) – For issue classification and prioritization

6. System Overview, Architecture, and Impact

This project introduces a centralized, real-time digital governance platform that connects citizens and municipal authorities in a secure and transparent system for managing public service complaints.

Citizen Portal

Citizens will be able to:

- Register and log in securely
- Submit complaints with descriptions, photos, and geo-location
- Track complaint status in real time
- Receive SMS or email notifications
- View updates on resolution progress
- Provide feedback after issue resolution

Municipal Dashboard

Authorities will be able to:

- View and filter issues by category and urgency
- Assign tasks to relevant departments
- Update issue progress
- Monitor department performance
- Analyze issue distribution by location
- Generate performance and compliance reports

AI & Smart Features

- Automatic issue categorization using natural language processing
- Detection of urgent cases (e.g., major water leakage)
- Hotspot analysis for frequently reported problems
- Predictive insights for high-risk areas
- Intelligent suggestions for resource allocation

7.Expected Impact

The system is expected to:

- Reduce response time for community issues
- Improve transparency and accountability
- Increase citizen participation and engagement
- Support data-driven municipal planning
- Strengthen public trust in local governance
- Improve operational efficiency within municipal departments

By digitizing the complaint management process and integrating intelligent decision support, the project contributes to smarter governance and more sustainable urban management in Ethiopia.

TITLE 3: An AI-Powered Legal Assistant for Ethiopian Labor and Proclamation Advisory

1. Introduction

Access to legal information is still a major challenge in Ethiopia, especially in areas related to labor rights and national proclamations. Although labor laws are established and supervised by institutions such as the Ethiopian Ministry of Labor and Skills and the Federal Supreme Court of Ethiopia, most legal documents are written in complex legal language that is difficult for ordinary citizens to understand.

Because of this, many employees are not fully aware of their rights regarding employment contracts, termination procedures, overtime payments, maternity leave, and workplace protections. On the other hand, employers and small business owners may unintentionally violate

labor regulations due to a lack of clear and accessible guidance. Professional legal consultation services are often expensive and not easily accessible to many citizens.

This project proposes an AI-powered legal advisory platform that aims to simplify Ethiopian labor proclamations and make them easier to understand. The system will allow users to ask legal questions in simple language and receive clear explanations, along with references to the relevant articles of the proclamation.

2. Problem Statement

The current legal information environment in Ethiopia faces several important challenges:

- **Limited Public Understanding of Labor Laws:** Legal proclamations are written in technical language that is not easily understood by non-legal professionals.
- **High Cost of Legal Consultation:** Many citizens cannot afford professional legal advice for basic legal questions.
- **Lack of Digital Legal Advisory Tools:** There is no centralized intelligent system that provides instant legal guidance based specifically on Ethiopian labor laws.
- **Difficulty Accessing Legal References:** Users must manually search through long and complex legal documents to find relevant information.
- **Risk of Misinformation:** People often rely on informal advice from unverified sources, which may lead to incorrect decisions.
- **Low Legal Awareness:** Both employees and employers may lack proper awareness of their rights and responsibilities.

These challenges contribute to misunderstandings, workplace disputes, and reduced compliance with labor regulations.

3. Project Objectives

General Objective

To design and implement a cloud-based AI-powered legal advisory system focused on Ethiopian Labor Proclamations that provides simplified explanations and practical legal guidance for citizens, employees, and employers.

Specific Objectives

The system aims to:

- Digitize and organize Ethiopian labor proclamations into a structured and searchable database.
- Allow users to ask legal questions in natural language (Amharic and/or English).
- Implement intelligent question analysis and response generation.
- Provide simplified explanations of complex legal articles.

- Display references to relevant proclamation sections for transparency.
- Develop role-based dashboards for administrators and legal supervisors.
- Securely store user queries for system improvement and analysis.
- Improve public legal awareness and support better regulatory insight.

4. Proposed System Overview

The proposed solution is an AI-enhanced legal advisory platform designed to make Ethiopian labor laws more accessible and easier to understand.

The system will serve three main user groups:

- Citizens and Employees (Public Portal)
- Employers and Business Owners
- System Administrators / Legal Supervisors

The intelligent component of the system will:

- Analyze user questions using Natural Language Processing (NLP).
- Identify relevant labor proclamation articles.
- Generate simplified and understandable explanations.
- Provide contextual legal guidance.
- Suggest related articles for further reading.

Role-based access control will be implemented to ensure secure and organized system usage.

5. Technology Stack

Frontend

- React.js – For building a dynamic and responsive user interface
- Tailwind CSS – For modern and accessible design
- Redux Toolkit – For managing application state efficiently
- React Router – For smooth navigation between pages
- Axios – For communication with the backend APIs
- Chart.js or Recharts – For displaying analytics and statistics

Backend

- Node.js with Express.js – For building RESTful APIs
- MongoDB – For storing structured legal articles and user queries
- Mongoose – For data modeling and validation
- JWT – For secure authentication and role-based authorization
- Socket.io (optional) – For real-time updates
- AI/NLP Integration – For question classification and intelligent response generation

6. System Overview, Architecture, and Impact

This project introduces a centralized digital legal advisory system that helps bridge the gap between complex labor proclamations and public understanding.

Public Legal Advisory Portal

Users will be able to:

- Ask legal questions in simple language
- Receive clear and simplified explanations
- View references to related proclamation articles
- Access frequently asked legal questions
- Save or export responses for future use

Employer & Business Module

Employers can:

- Review compliance guidelines
- Understand legal obligations related to employee rights and contracts
- Access structured summaries of labor regulations

Administrative Dashboard

Administrators will:

- Manage and update labor proclamation content
- Monitor user queries and usage statistics
- Review analytics for system improvement
- Ensure accuracy and proper system performance

7. Advanced Features and Technology

The system will include:

- NLP-based question understanding
- Intelligent retrieval of relevant legal articles
- Simplified explanation generation
- Multilingual support (Amharic and English)
- Legal disclaimers to clarify advisory limitations
- Usage analytics for better insight

The system will be deployed on a secure cloud environment with encrypted communication to ensure data protection.

8. Expected Impact

The proposed system is expected to:

- Improve public understanding of labor rights and responsibilities
- Reduce misinformation and workplace disputes
- Increase compliance with labor regulations
- Promote transparency and legal awareness
- Reduce dependence on expensive legal consultations for basic inquiries
- Support digital transformation within Ethiopia's legal information sector